Basic Power BI Interview Questions

1. What is Power BI?

Answer:
Power BI is a business intelligence and data visualization tool developed by Microsoft. It allows users to connect to various data sources, transform data, create reports, and share insights through dashboards and visualizations.

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2. What are the key components of Power BI?

Answer:
The main components of Power BI are:

- **Power BI Desktop** – A tool for report creation.
- **Power BI Service (Cloud)** – A cloud-based service for sharing reports.
- **Power BI Mobile** – A mobile app for accessing reports.
- **Power Query** – A data transformation engine.
- **Power Pivot** – A data modeling engine.
- **Power View** – A data visualization tool.
- **Power BI Gateway** – A bridge for on-premises data.

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3. What data sources can Power BI connect to?

Answer:
Power BI can connect to multiple data sources, including:

- Excel
- SQL Server
- Azure
- Oracle
- Google Analytics
- Salesforce
- Web APIs
- SharePoint
- CSV and JSON files

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4. What is the difference between Power BI Desktop and Power BI Service?

Answer:

- **Power BI Desktop** is a local application used to create reports and dashboards.
- **Power BI Service** is a cloud-based platform used for publishing, sharing, and collaboration.

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5. What is a Power BI Dashboard?

Answer:
A **dashboard** is a single-page, interactive visualization that consolidates data from multiple reports. It helps in monitoring key performance indicators (KPIs) at a glance.

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6. What is a dataset in Power BI?

Answer:
A dataset is a collection of data imported or connected in Power BI. It serves as the foundation for creating reports and visualizations.

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7. What is DAX in Power BI?

Answer:
DAX (Data Analysis Expressions) is a formula language used in Power BI for calculations and data analysis. It is used in calculated columns, measures, and calculated tables.

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8. What is a Power BI Report?

Answer:
A **Power BI Report** is a multi-page collection of visualizations and insights based on data models.

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9. What is Row-Level Security (RLS) in Power BI?

Answer:
RLS allows you to restrict data access at the row level based on user roles. It is implemented using DAX filters in Power BI.

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10. What is the difference between Import Mode and DirectQuery?

Answer:

- **Import Mode**: Data is stored in Power BI, allowing fast performance.
- **DirectQuery**: Queries are executed live on the source system, reducing data storage needs.

Intermediate Power BI Interview Questions

11. What is Power Query?

Answer:
Power Query is a tool used in Power BI for **Extract, Transform, and Load (ETL)** operations. It helps in cleaning and transforming data before loading it into the model.

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12. What are the different types of filters in Power BI?

Answer:

- **Visual-level filters** – Apply to a specific visualization.
- **Page-level filters** – Apply to an entire report page.
- **Report-level filters** – Apply to the whole report.
- **Drillthrough filters** – Focus on a specific report section.

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13. What is the difference between a calculated column and a measure in Power BI?

Answer:

- **Calculated Column**: Stored at the row level in the table.
- **Measure**: Computed dynamically based on user interactions.

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14. What is the use of the CALCULATE() function in DAX?

Answer:
The CALCULATE() function modifies the context of a calculation. Example:

CALCULATE(SUM(Sales[Amount]), Sales[Region] = "West")

It calculates total sales only for the "West" region.

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15. What is a star schema in Power BI?

Answer:
A **star schema** consists of a **fact table** (central table with numeric data) linked to **dimension tables** (descriptive attributes).

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16. How do you handle performance issues in Power BI?

Answer:

- Use **Import Mode** instead of DirectQuery.
- Reduce the number of columns in datasets.
- Use **aggregated tables**.
- Optimize DAX calculations.

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17. What is the use of the SUMX() function in DAX?

Answer:
SUMX() performs calculations row by row. Example:

SUMX(Sales, Sales[Quantity] * Sales[UnitPrice])

Calculates total sales by multiplying quantity and price per row.

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18. How does Power BI handle real-time data?

Answer:
Power BI supports real-time data through:

- Streaming datasets
- Push datasets
- DirectQuery mode

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19. What is a Power BI Gateway?

Answer:
A **Gateway** acts as a bridge between **on-premises** data sources and the Power BI Service.

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20. What is drill-through in Power BI?

Answer:
Drill-through allows users to navigate from **summary-level** data to **detailed-level** data by clicking on a visualization.

Advanced Power BI Interview Questions

21. What is a composite model in Power BI?

Answer:
A **composite model** allows combining **Import Mode and DirectQuery** in a single dataset.

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22. What is Incremental Refresh in Power BI?

Answer:
Incremental Refresh allows updating only the **new or changed data** instead of refreshing the entire dataset.

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23. What is the difference between SUM() and SUMX()?

Answer:

- SUM(): Aggregates a column's total.
- SUMX(): Iterates row by row before summing.

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24. How do you use Python or R in Power BI?

Answer:
Python and R can be used in **Power Query Editor** for **data transformation** and in **visualizations** for **advanced analytics**.

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25. What is Power BI Paginated Reports?

Answer:
Paginated Reports are pixel-perfect reports designed for printing or exporting.

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26. How do you optimize Power BI performance?

Answer:

- Use **proper indexing** in databases.
- Optimize **DAX queries**.
- Reduce **cardinality** in datasets.
- Use **aggregated tables**.

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27. How do you implement a dynamic title in Power BI?

Answer:
Use a **DAX measure**:

TitleMeasure = "Sales Report for " & SELECTEDVALUE(Sales[Region])

Then bind it to a card visualization.

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28. What is the use of the ALLEXCEPT() function?

Answer:
ALLEXCEPT() removes all filters **except** the specified ones.

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29. What are Power BI Dataflows?

Answer:
Power BI Dataflows store ETL-transformed data in the cloud using Azure Data Lake.

30. What are parameters in Power BI?

Answer:
Parameters allow users to **dynamically control** queries, filters, and calculations.